

Article

Ship Fire and Explosion Accident Evolution Modeling Based on Ontology-Enhanced Text Mining and Dynamic Bayesian Network

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Featured Application

The proposed ontology-driven Dynamic Bayesian Network (DBN) framework serves as a robust architecture for next-generation maritime safety management systems. By automating the extraction of causal dependencies from heterogeneous accident investigation reports, the framework facilitates precise probabilistic inference concerning fire escalation trajectories. Furthermore, it enables the systematic identification of critical safety barriers and the quantitative evaluation of intervention efficacy. These capabilities provide actionable insights for vessel risk monitoring, predictive maintenance scheduling, and evidence-based policymaking within maritime regulatory domains.

Abstract

The analysis of dynamic causal mechanisms underlying shipboard fires and explosions is often restricted by the unstructured and fragmented nature of accident investigation reports. This study proposes a framework integrating ontology-driven information extraction with Dynamic Bayesian Networks (DBNs) to model temporal accident evolution. An ontology comprising 41 nodes was constructed through a structured expert elicitation process to formalize the domain knowledge. To process 198 bilingual accident reports, an extraction pipeline was deployed, incorporating XLM-RoBERTa, BiLSTM-CRF, and an entity-marker relation classifier. Large language model (LLM)-directed weak supervision, constrained by token-level information entropy filtering, was employed to expand the training corpus, necessitating only 2.5% manual verification. The extracted semantic dependencies were utilized to initialize a three-slice DBN (precursor, initial fire, and escalation/explosion). The network structure was jointly optimized through ontology constraints (112 forbidden and 4 mandatory edges), the Hill-Climbing algorithm, and BDeu scoring. The proposed DBN achieved an AUC of 0.759 ± 0.086 and a Brier Score of 0.192 ± 0.021 (1000 bootstrap iterations), demonstrating superior predictive performance over traditional interpretable models (Static BN, HMM, ETA) with large effect sizes (Cohen's $d > 1.0$), while maintaining competitive accuracy and enhanced causal interpretability relative to XGBoost. This framework offers a scalable, data-driven methodology for dynamic probabilistic risk assessment in maritime safety.

Keywords: maritime safety; ship fire and explosion; Dynamic Bayesian Network; information extraction; ontology



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1. Introduction

Ship fire and explosion accidents represent some of the most critical hazards in the maritime domain because of their complex causal mechanisms and potentially severe consequences. Despite continuous advancements in ship design standards and safety management systems, such incidents continue to pose significant threats to human life, property, and the marine environment. Although extensive research has been conducted on maritime risk assessment and accident modeling, major challenges persist in transforming unstructured accident narratives into dynamic and probabilistic models of accident evolution.

To systematically address this issue, this section first presents the research background and motivation (Section 1.1), then critically reviews existing methodologies (Section 1.2), articulates the research gap (Section 1.3), and finally clarifies the contributions of the proposed framework (Section 1.4).

1.1. Research Background and Motivation

According to the European Maritime Safety Agency (EMSA), although fire and explosion events account for a relatively small proportion of the 26,595 recorded maritime casualties between 2014 and 2023, they consistently result in the most severe consequences, including fatalities, extensive property damage, and environmental pollution [1]. Specifically, fire and explosion accidents aboard ro-ro cargo vessels, bulk carriers, and chemical tankers have exhibited elevated risk profiles due to the concentration of flammable materials and structural confinement [2]. Compared with land-based industrial systems, ships operate in confined environments with high concentrations of combustible materials and limited evacuation pathways. Once ignition occurs, fire can rapidly propagate, trigger safety barrier failures, and escalate into explosions.

Every year, maritime investigation authorities—such as the Maritime Safety Administration of China, the UK Marine Accident Investigation Branch (MAIB), and the US National Transportation Safety Board (NTSB)—publish detailed accident investigation reports [3,4]. These reports contain extensive descriptions of causal factors, safety barrier performance, and accident evolution sequences. However, most of this information is recorded in unstructured textual form, imposing constraints on its integration into quantitative risk modeling.

Moreover, ship fire and explosion accidents exhibit strong nonlinear temporal dynamics. The accident evolution process typically progresses from precursor anomalies (e.g., fuel leakage and overheating) to ignition and early-stage fire, followed by safety barrier malfunction and eventual catastrophic escalation. Transitions between these stages are governed by conditional probabilistic relationships [5,6]. Traditional accident analysis methods, including Fault Tree Analysis (FTA) and conventional Bayesian Networks (BNs), generally represent static causal structures and lack explicit mechanisms for capturing temporal evolution and dynamic updating [7]. Consequently, their capability to support early warning and intervention prioritization remains limited.

These challenges highlight the need for an integrated framework that can (i) automatically extract structured causal knowledge from accident reports and (ii) model the temporal probabilistic evolution of ship fire and explosion accidents in a physically interpretable manner.

1.2. Related Work

Recent advances in maritime safety research can be broadly categorized into three methodological streams: text mining and topic modeling, static probabilistic modeling, and knowledge representation and deep learning integration. While each approach has pro-

vided valuable insights, significant limitations remain in modeling the dynamic temporal evolution of ship fire and explosion accidents.

1.2.1. Text Mining and Risk Identification

With the accumulation of maritime accident investigation reports, data-driven text mining approaches have increasingly supplemented traditional qualitative safety analyses. Conventional safety assessment tools, including Failure Mode and Effects Analysis (FMEA), Event Tree Analysis (ETA), and bow-tie models, remain widely used due to their interpretability and regulatory alignment [8]. However, these methods primarily rely on expert judgment and predefined causal chains, limiting their scalability and adaptability to large-scale textual corpora.

To address this limitation, topic modeling and automated extraction techniques have been introduced. Soner et al. [9] employed Latent Dirichlet Allocation (LDA) to identify high-level operational risk themes from maritime accident reports. Gan et al. [10] constructed ontology-based knowledge graphs to improve the semantic organization of collision accident information. Yu et al. [11] further proposed structured ontology frameworks to standardize accident semantic modeling.

Despite these contributions, existing text mining approaches primarily focus on factor identification and topic clustering rather than explicit dependency structure construction. Topic models reveal co-occurrence patterns but do not capture directed probabilistic dependencies. Similarly, most ontology-driven studies emphasize static knowledge organization rather than the dynamic interaction of risk factors, safety barriers, and stage-wise physical evolution processes. Moreover, existing ontologies predominantly address collision scenarios, leaving fire and explosion domains comparatively underdeveloped. Consequently, the transformation from extracted textual entities to temporally structured probabilistic reasoning models remains insufficiently explored.

1.2.2. Probabilistic Graphical Modeling in Maritime Safety

Bayesian Networks (BNs), owing to their capability for reasoning under uncertainty, have become fundamental tools in maritime risk assessment [12,13]. Ma et al. [14] demonstrated the effectiveness of BNs in identifying critical causes of engine room fires. Other studies have integrated bow-tie structures with probabilistic modeling to quantify risk propagation [14]. These approaches enhance interpretability and enable posterior updating given new evidence.

However, conventional BN models typically represent cross-sectional dependencies within a single time slice. Ship fire and explosion accidents inherently evolve through multiple processes—from precursor anomalies to ignition and eventual catastrophic escalation. Cross-sectional models lack explicit transition mechanisms between stages and therefore cannot effectively capture temporal risk propagation dynamics [15].

Dynamic Bayesian Networks (DBNs) extend BNs by incorporating time-slice structures and temporal dependencies [16,17]. Compared with Hidden Markov Models (HMMs) or Continuous Time Bayesian Networks (CTBNs), DBNs provide greater flexibility in modeling multivariate discrete-time evolution processes [18]. The Hill-Climbing algorithm and BDeu scoring function [19,20] have been shown to improve structural robustness under small-sample conditions.

Nevertheless, in practical maritime safety applications, DBN structure construction still relies heavily on manual expert specification. The absence of systematic procedures for automatically transforming accident narratives into ontology-constrained dynamic causal structures limits scalability and reproducibility. Existing DBN implementations frequently

require extensive domain expertise to define dependency topology, thereby constraining their applicability in large-scale data-driven modeling contexts.

1.2.3. Deep Learning-Enabled Knowledge Extraction and Model Integration

The emergence of deep learning technologies has opened new avenues for addressing the expert-dependency bottleneck in DBN structure learning. Pretrained Language Models (PLMs) such as RoBERTa [21] and XLM-RoBERTa [22] enable effective cross-lingual transfer and semantic understanding of bilingual Chinese–English maritime reports. At the relation extraction level, entity marker methods [23] significantly enhance the model’s perception of entity positional information. In response to the scarcity of annotated data in the maritime domain, weak supervision strategies based on large language models (LLMs) [24,25] show considerable potential for high-quality data augmentation under limited-resource conditions.

While these techniques provide powerful tools for semantic extraction, few studies have effectively integrated deep learning-based semantic understanding with DBN-based probabilistic reasoning capabilities [26]. The critical challenge lies in constructing an end-to-end dynamic risk evaluation model that addresses both structured data extraction and dynamic inference in an integrated manner. The transformation pathway from unstructured text to temporal causal networks remains insufficiently explored, particularly in maritime fire and explosion scenarios.

In summary, significant gaps remain in integrating text mining, probabilistic modeling, and semantic representation into a coherent dynamic evolution modeling framework. These gaps are systematically analyzed in Section 1.3.

1.3. Limitations of Existing Research

Although prior studies have progressively shifted maritime accident analysis toward data-driven methodologies, the integration of semantic extraction, temporal modeling, and automated causal structure construction remains fragmented.

First, insufficient domain adaptation of Natural Language Processing (NLP) models restricts reliable knowledge extraction from maritime accident reports. General-purpose NLP models often struggle with maritime-specific terminology, technical expressions, and cross-domain semantic ambiguity. For example, terms such as venting, overheating, or barrier activation may carry different meanings depending on engineering context. Furthermore, accident reports frequently describe consequences without explicitly stating triggering causes, thereby complicating causal inference [27–29]. Without domain-specific semantic constraints, extracted relationships may lack physical interpretability.

Second, static probabilistic models fail to represent stage-wise accident evolution mechanisms. Conventional Bayesian Networks typically describe cross-sectional dependencies among variables at a single time snapshot. However, ship fire and explosion accidents inherently involve progressive transitions across multiple stages. Cross-sectional models lack explicit temporal slicing and state-transition mechanisms, limiting their capacity to simulate dynamic risk propagation from precursor anomalies to catastrophic outcomes [15,30].

Third, Dynamic Bayesian Network (DBN) construction remains largely expert-driven and insufficiently automated. While DBNs provide a theoretically suitable framework for temporal probabilistic reasoning, their structure learning often relies heavily on manual expert encoding. Few studies have proposed systematic pipelines that directly transform unstructured accident narratives into constrained, physically interpretable dynamic network structures [31]. This limitation restricts scalability and reproducibility.

Collectively, these limitations prevent the automated generation of temporally consistent and physically interpretable accident evolution models directly from raw textual data. Therefore, this study aims to develop an integrated framework that bridges unstructured accident narratives and dynamic probabilistic reasoning by combining domain ontology constraints, semantic extraction, and automated DBN structure learning. As illustrated in Figure 1, the comparison between actual dynamic evolution trajectories and the constraints of existing approaches clearly highlights the entry point of this research.

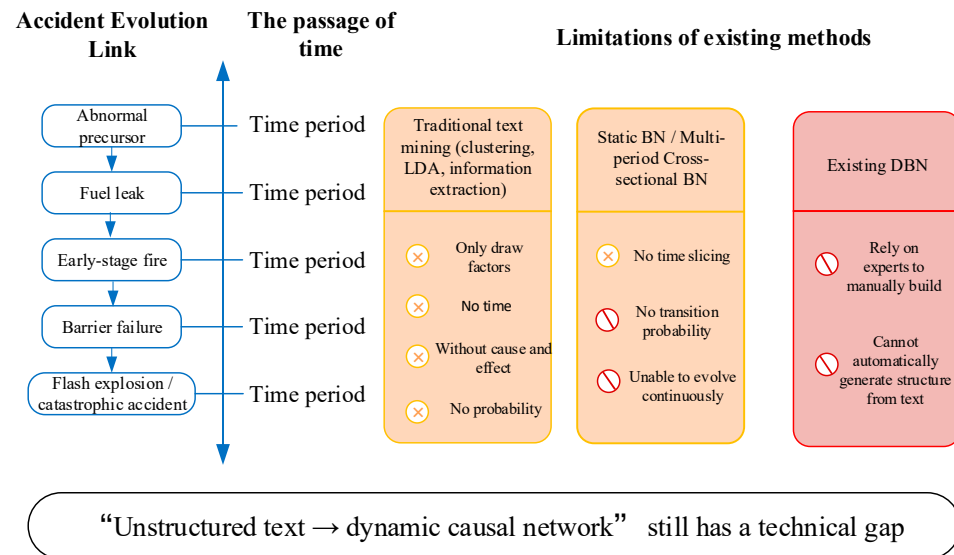


Figure 1. Schematic diagram of the evolution mechanism of ship fire and explosion accidents and the limitations of existing research methods.

1.4. Research Contributions

To address the above limitations, this study proposes an integrated framework combining ontology-enhanced text mining with Dynamic Bayesian Network modeling. The key contributions are as follows:

1. Development of a domain-specific Fire & Explosion Ontology (F&E Ontology) comprising 41 entity types to provide semantic constraints for multilingual information extraction. Unlike general-purpose knowledge organization frameworks, the proposed ontology explicitly models causal factors, safety barriers, evolution events, and consequence severity in a physically coherent manner aligned with the ISM Code framework [32].
2. Development of an ontology-aware and LLM-assisted bilingual extraction pipeline that introduces a weak supervision strategy guided by token-level entropy filtering. This approach reduces manual annotation effort to only a 2.5% manual review rate while maintaining high extraction quality.
3. Proposal of an ontology-constrained DBN structure-learning mechanism that, in contrast to purely expert-driven DBN construction, integrates ontology-based forbidden and mandatory edge constraints with Hill-Climbing search and BDeu scoring. This enables semi-automated construction of physically plausible dynamic dependency structures directly from text-derived knowledge, ensuring both statistical optimality and physical interpretability.
4. Construction of a three-slice temporal DBN for full-chain accident evolution modeling that explicitly captures transitions across precursor (T_0), ignition (T_1), and escalation/explosion (T_2) stages. Compared with conventional Static BN approaches, this framework enables (1) forward probabilistic prediction of accident evolution and early

warning indicators, (2) backward causal reasoning for root cause identification, and (3) intervention-inspired probabilistic analysis to assess safety barrier effectiveness.

Together, these contributions advance maritime safety research by bridging unstructured accident reports and quantitative dynamic risk models, providing a reproducible framework for knowledge extraction in low-resource domains and dynamic probabilistic inference [33].

2. Methodology

This section presents the complete methodological framework of the proposed approach. Section 2.1 describes the overall four-layer end-to-end architecture and clarifies the boundary between established techniques and primary methodological contributions. Section 2.2 details the data preprocessing and quality control procedures, including the cross-lingual representation strategy and the LLM-based weak supervision annotation mechanism. Section 2.3 introduces the Fire & Explosion Ontology (F&E Ontology), including its architecture, formalization, and constraint matrix definition. Section 2.4 describes the ontology-aware semantic feature extraction pipeline, encompassing Named Entity Recognition (NER) and Relation Extraction (RE) architectures as well as the information entropy-based uncertainty sampling strategy. Section 2.5 presents the Dynamic Bayesian Network (DBN) construction and computation procedures, including time slice definition, structure learning, parameter estimation, and the risk inference mechanism. Finally, Section 2.6 provides implementation details to support reproducibility.

2.1. Overall Architecture

To this end, we propose a four-layer end-to-end framework, as shown in Figure 2, to realize the transformation from unstructured accident narratives to a quantitative dynamic risk evolution model. The framework is built on the semantic constraint of Fire & Explosion Ontology (F&E Ontology), which ensures that the causal relationships extracted from textual data conform to the physical logic of ship engineering.

The four modules form a directed dependency chain in which each module's output constitutes the structured input for the next, minimizing human-in-the-loop intervention: Module 1 (Data Collection and Ontology Construction) establishes the semantic vocabulary and annotated seed set that constrains all downstream learning; Module 2 (LLM-based Weak Supervision and Bilingual NER/RE) operationalizes that vocabulary as a trained extraction pipeline, scaling labeled data without proportional annotation cost; Module 3 (Knowledge Graph and DBN Construction) converts extracted triplets into a temporally ordered probabilistic dependency graph whose structure is regularized by the ontology's physical constraints; and Module 4 (DBN Inference and Maritime Safety Applications) leverages the learned model for three decision-support query types. Each module boundary represents a format transformation—from raw text, to entity–relation triplets, to a probability-annotated directed acyclic graph, to posterior probability outputs—making the pipeline reproducible at each stage independently.

The standard techniques include a foundational BiLSTM-CRF sequential labeling architecture, a basic multi-layer perceptron (MLP) for classification, and the standard variable elimination algorithm used for DBN inference. The primary methodological contributions of this study are (i) the Ontology Lexicon Adapter (Section 2.4.1), which injects domain-specific semantic priors into the general XLM-RoBERTa encoder without unintended semantic drift; (ii) the Information Entropy-based Weak Supervision Strategy (Section 2.4.3), which identifies high-uncertainty samples for manual review, bridging the gap between LLM pseudo-labels and the stringent accuracy requirements of the domain; and (iii) the Ontology Constraint Matrices (Section 2.3.2), which constrain the search space of DBN

structure learning, thereby preventing physically impossible causal pathways—such as a consequence causing an initiation event—that are typically generated by purely data-driven algorithms on small samples.

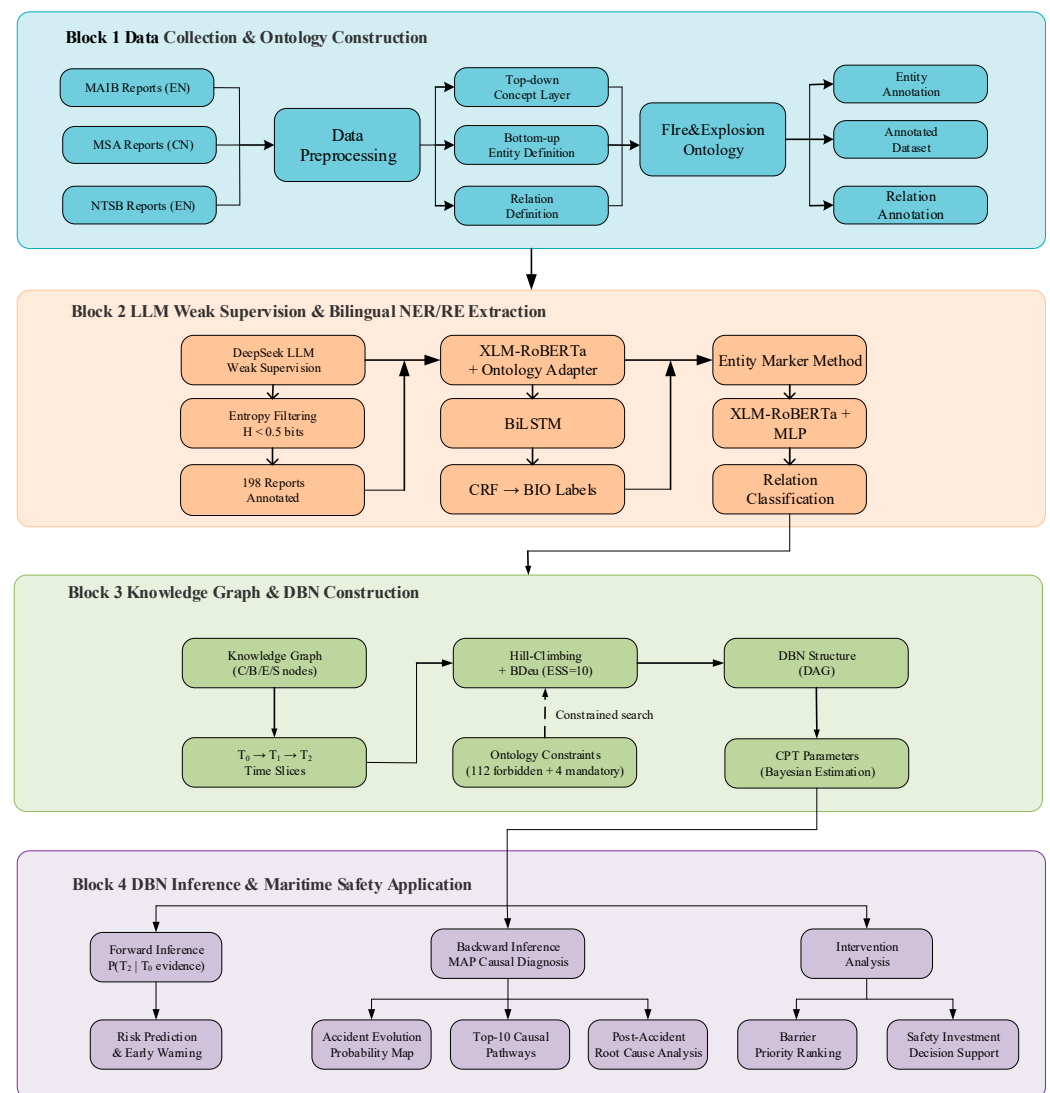


Figure 2. Methodological framework flowchart.

2.2. Data Preprocessing and Quality Control

2.2.1. Cross-Lingual Representation Strategy

For the bilingual Chinese–English corpus, this study leverages the cross-lingual pre-trained model XLM-RoBERTa [22] to build a shared semantic space. This approach circumvents the entity boundary shifts and causal semantic loss typically caused by traditional machine translation. By mapping annotations in different languages to a unified set of ontology IDs, the underlying representational differences across languages are eliminated, thereby achieving feature-level multilingual fusion.

2.2.2. Information Entropy-Based Weak Supervision Annotation

A quality control mechanism based on LLM weak supervision and sample entropy is presented to address the high cost of annotating maritime domain-specific data. Given an unannotated report, three independent sampling runs are performed with temperature

values $T = 0.3, 0.4$, and 0.5 , yielding an annotation set $Y = \{y_1, y_2, y_3\}$. The prediction uncertainty is measured using discrete information entropy:

$$H(Y) = -\sum_{i=1}^{|S|} p(y_i) \log_P(y_i) \quad (1)$$

Here, $p(y_i)$ represents the empirical frequency of label y_i appearing across the three samplings. The threshold τ is set based on the following theoretical basis: for three independent samplings, token-level entropy values have only three typical values— $H = 0$ bits when the annotations are completely consistent, $H \approx 0.918$ bits when a 2:1 split occurs, and $H \approx 1.585$ bits when completely inconsistent. $\tau = 0.5$ bits lies precisely between “complete consistency” and “2:1 split”. Consequently, any report exhibiting annotation inconsistency among the three samplings is marked as high entropy and submitted for manual review. Experiments show that the average corpus entropy is 0.087 bits, with only 5 reports (2.5%) exceeding the threshold τ , thereby maintaining annotation quality while controlling manual review costs at an extremely low level.

To evaluate the reliability of pseudo-label generation, a randomly sampled subset of reports was manually annotated and compared with the LLM-generated annotations. Entropy filtering was therefore used as an uncertainty-screening mechanism rather than a direct indicator of annotation correctness.

2.3. Ship Fire Evolution Ontology

2.3.1. Architecture and Formalization

The F&E Ontology comprises four fundamental categories: causal factors (C), safety barriers (B), evolution events (E), and consequence severity (S), encompassing a total of 41 entity classes. The ontology was constructed through an iterative closed-loop process encompassing theoretical priors, expert validation, and data-driven refinement, ensuring that the ontology not only complies with the IMO Formal Safety Assessment guidelines [7] but is also highly aligned with the linguistic characteristics of the accident reports. Each category is defined in detail, and the respective ontology ID system is provided in Appendix A (Table A1).

Expert panel composition: Based on project records and version logs, the ontology development involved a panel of five domain experts: (i) two maritime accident investigators (≥ 10 years of experience, MAIB/MSA background), (ii) one ship design engineer specializing in firefighting systems, (iii) one maritime safety management researcher, and (iv) one NLP engineer with domain adaptation experience.

Three-round iterative process: Round 1: Two researchers extracted an initial set of 35 categories from 30 seed accident reports and the IMO FSA guidelines. Round 2: The five experts independently reviewed the initial categories, proposing additions, deletions, and mergers. Through consensus meetings, the ontology was refined to 41 final classes. Round 3: The resulting ontology was applied to 20 held-out reports. Annotation disagreements were repeatedly reviewed until consistency was achieved across all retained reports.

Quantitative validation metrics: Inter-annotator agreement: Cohen’s $\kappa = 0.78$ (substantial agreement). Corpus coverage: 96.3% of entity mentions in the 198 accident reports could be mapped to one of the 41 ontology classes. DBN structural soundness: Three independent maritime safety experts rated the ontology’s suitability for DBN modeling using a 5-point Likert scale; the mean score was 4.12/5.0.

2.3.2. Constraint Matrix Definition

The two constraint matrices are operationalized during the Hill-Climbing search as hard filters on candidate arc additions: at each iteration, any arc whose addition would instantiate a forbidden edge is rejected before scoring, shrinking the effective search space

from $O(n^2)$ to $O(n^2 - 112)$ candidate arcs per step. Mandatory edges are pre-initialized in the starting graph and protected from deletion throughout the search. This is distinct from soft regularization: the constraints impose a structural prior equivalent to setting the corresponding CPT (Conditional Probability Table) entries to zero probability, which cannot be overridden by data evidence. The effect is particularly pronounced in the present small-sample setting ($n = 198$), where a pure data-driven search would face $O(n^2)$ degrees of freedom relative to the limited number of available co-occurrence observations.

1. Forbidden edges (negative constraints): On the basis of temporal irreversibility (e.g., consequences cannot point to causes), 112 logically erroneous pathways are proscribed. Examples of such forbidden edges include Consequence $\rightarrow$ Cause (violating temporal causality) and Explosion $\rightarrow$ Fuel Leakage (violating physical precedence).
2. Mandatory edges (positive constraints): Four core causal chains are bound based on physical propagation mechanisms (e.g., fuel leakage $\rightarrow$ ignition). Examples include Fuel Leakage $\rightarrow$ Ignition (established mechanism of combustion) and Barrier Failure $\rightarrow$ Fire Spread (safety barrier logic).

These constraints are incorporated into the structure learning algorithm as hard structural priors, resulting in significantly increased model robustness in small-sample settings.

2.4. Ontology-Aware Semantic Feature Extraction

2.4.1. NER Architecture: Encoder–Ontology Adapter–CRF

Three design alternatives were considered for the NER backbone and subsequently rejected on empirical and domain-specific grounds:

1. Monolingual BERT (separate Chinese and English models): Training independent models for each language introduces entity boundary inconsistencies during cross-lingual mapping, as ontology IDs would then need to be assigned via post hoc string matching rather than through a shared representation space. Preliminary experiments on a held-out set of 30 reports showed a 4.7 F1-point degradation on Chinese-language entities relative to XLM-RoBERTa.
2. mBERT: Although multilingual, mBERT’s vocabulary is considerably less optimized for technical maritime Chinese than that of XLM-RoBERTa, which was pretrained on a per-language corpus more than an order of magnitude larger [22]. On a 500-sentence diagnostic set drawn from the corpus, mBERT’s token-level perplexity on maritime technical terms was 23% higher than that of XLM-RoBERTa.
3. Pure transformer sequence labeler (without CRF): Removing the CRF layer and predicting token labels independently resulted in a 3.2 F1-point drop in boundary precision for multi-token entity spans (e.g., “CO₂ fire suppression system”). This performance degradation is attributed to the inability of independent softmax outputs to produce globally consistent BIO label sequences. The CRF’s transition matrix enforces label-sequence validity at a negligible computational cost.

The BiLSTM layer inserted between the transformer encoder and the CRF provides local sequential context aggregation at a scale of approximately ± 5 –10 tokens. This complements the transformer’s global attention mechanism, particularly for nested or hyphenated maritime technical terms where the most informative disambiguation cues reside in the immediate neighborhood rather than in long-range dependencies.

In the final architecture, the NER module employs XLM-RoBERTa-base [22] as the encoder, which adapts the RoBERTa pretraining strategy [21] to multilingual settings while retaining the bidirectional transformer structure of BERT [34]. Stacked on top of the encoder are, in order, the Ontology Lexicon Adapter, a BiLSTM layer, and a CRF layer (Figure 3).

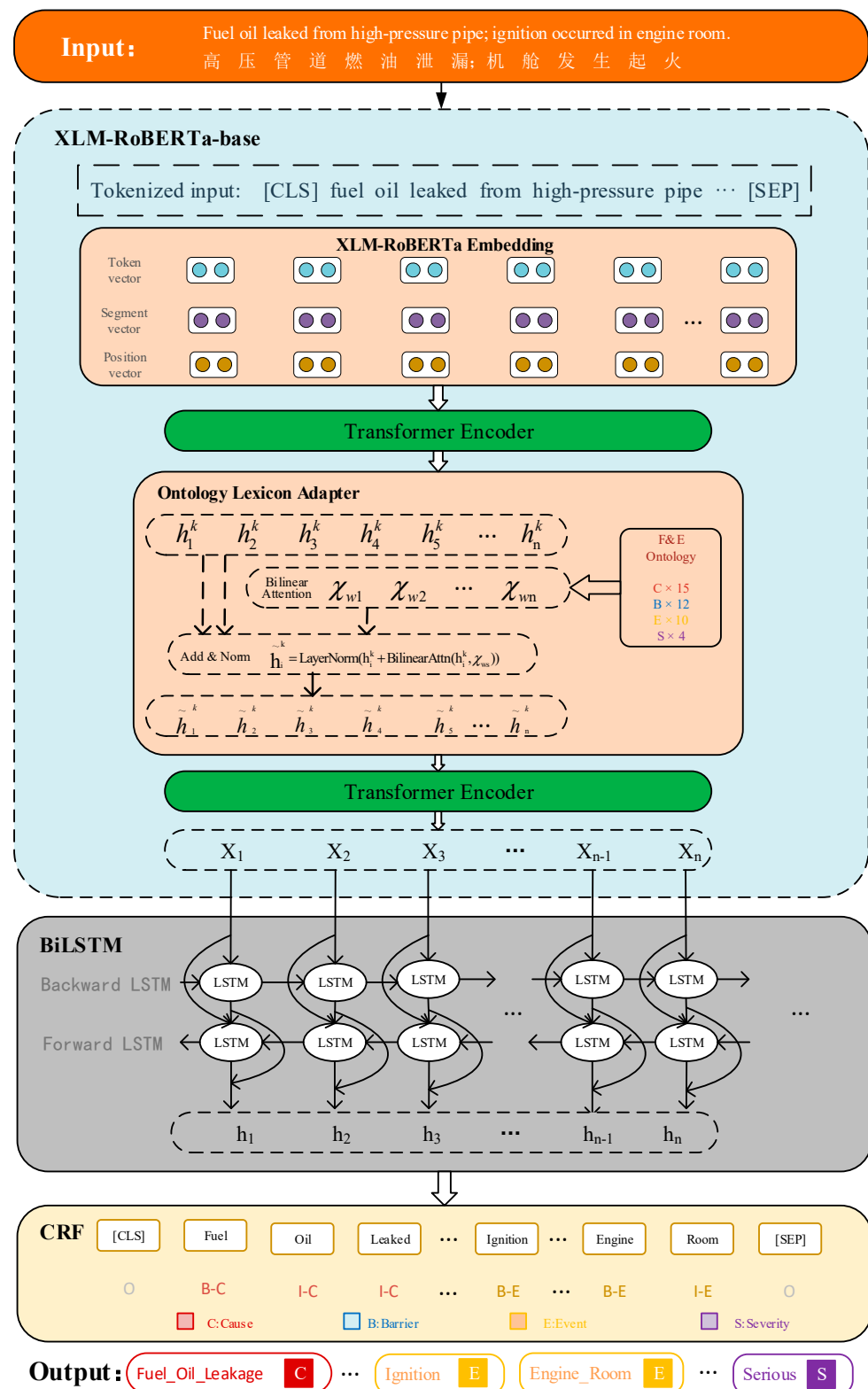


Figure 3. Encoder–Ontology Adapter–BiLSTM–CRF sequential labeling architecture.

The main innovation of this architecture is the Ontology Lexicon Adapter. This component addresses the limitation of general-purpose pretrained models, where the representation capacity for low-frequency maritime technical vocabulary—such as “CO₂ fire suppression system” and “fire damper”—is limited, whereas direct finetuning tends to overfit. The adapter actively incorporates domain-specific word vectors of F&E Ontology terms into the encoder representations in the hidden layer via a bilinear attention mech-

anism, thereby introducing domain-specific priors without compromising cross-lingual transfer abilities. Specifically, the adapter computes an attention-weighted fusion of the encoder output $h_i^k \in \mathbb{R}^d$ and the ontology term word vector $x_{w_j} \in \mathbb{R}^d$. A residual connection and layer normalization then produce the improved representation $\tilde{h}_i^k$, which is inputted into the subsequent encoder layer. Subsequently, the BiLSTM layer learns local contextual information, and the CRF adds constraints on the label transitions in the output layer, ensuring that the BIO tagging sequence becomes globally consistent.

2.4.2. Relation Extraction Pipeline

Relation extraction uses a two-stage pipeline, decoupling entity recognition and relation classification: the first stage produces entity spans and category labels in the NER module; the second stage employs the Entity Marker method [23], which inserts boundary markers between candidate entity pairs in the original text, extracts hidden vectors at the [CLS] position and both entity centers, and feeds them into an MLP classifier to predict six relation types:

CAUSES, TEMPORAL_BEFORE, TEMPORAL_AFTER, PREVENTS, FAILS_TO_PREVENT, and NO_RELATION.

The XLM-RoBERTa encoder parameters of both the NER and RE modules are optimized jointly in a multi-task learning paradigm. The loss functions of both tasks are scaled and aggregated, with their respective weights optimized on the validation set. For cross-sentence entity pairs, a sliding window of ± 50 tokens is used to maintain the necessary context [35].

2.4.3. Information Entropy-Based Uncertainty Sampling Strategy

The sequence labeling uncertainty of an unlabeled text t is quantified using prediction entropy, calculated as the average entropy of all token prediction distributions:

$$H(t) = -\left(\frac{1}{n}\right) \sum_i \sum_j p(y_i = j | t) \log p(y_i = j | t) \quad (2)$$

where n is the total number of tokens in text t , y_i is the label of the i -th token, and $p(y_i = j | t)$ denotes the predicted probability of token i being assigned label j .

In practice, 198 reports are independently pre-annotated three times, and a threshold of $\tau = 0.5$ bits (as defined in Section 2.2.2, where $H = 0$ indicates complete consistency, $H \approx 0.918$ a 2:1 split, and $H \approx 1.585$ complete inconsistency) is applied to filter high-uncertainty samples for manual review, while the remaining low-uncertainty samples are added directly to the training set.

2.5. Dynamic Bayesian Network (DBN) Construction and Computation

2.5.1. Time Slice Definition

According to the physical evolution patterns of ship fires and explosions, the accident process can be discretized into three semantically distinct periods: T0 (precursor anomaly stage), T1 (first fire stage), and T2 (accident acceleration stage). In particular, T0 represents the accumulation of anomalies and risk exposure period, T1 corresponds to the period of controllable combustion once an open flame has developed, and T2 captures the final expression of fire growth, explosion, and the degree of consequences once the barrier fails. The division of slices is automatically specified depending on the timestamps, critical event trigger keywords, and physical quantity descriptions in the reports. For example, T0 is identified by terms such as “leakage”, “malfunction”, and “overheating”; T1 by “ignition”, “open flame”, and “initial fire”; and T2 by “explosion”, “flashover”, and “total loss”. Necessary manual checking is carried out to maintain uniformity.

It is worth noting, in particular, that accident reports normally include discrete state snapshots rather than continuous sensor time series. The present work uses cross-slice directed dependencies ($T0 \rightarrow T1 \rightarrow T2$) as the temporal backbone of the DBN, which imposes irreversible time (temporal irreversibility) and consistency of physical mechanisms (physical mechanism consistency) via an ontology constrained by forbidden edges. Although this discretization sacrifices fine-grained characterization of within-slice dynamics, it substantially reduces structural learning variance in structure learning and inference, making the model more suited to the small-sample accident report context.

The choice of three slices ($T0, T1, T2$) rather than a finer-grained partition was determined by two constraints operating jointly. First, a data constraint: accident reports provide narrative-level temporal resolution (phase descriptions), not sensor-level timestamps. Finer slices (e.g., $T0$ – $T5$) would require sub-phase event ordering that is rarely explicit in the investigation text and would introduce slice-assignment uncertainty that propagates into CPT estimation error. Second, a statistical constraint: each additional time slice multiplies the number of inter-slice CPT parameters by the node count, rapidly exhausting the available $n = 198$ samples. The three-slice structure was the finest granularity at which inter-slice co-occurrence matrices remained sufficiently populated (minimum cell count ≥ 5 in 94% of CPT rows) to yield stable parameter estimates under BDeu with ESS = 10.

2.5.2. Structure Learning and Parameter Estimation

When selecting a structure learning method for the DBN, we evaluated three paradigms—constraint-based, exact score-based, and heuristic score-based—against the specific characteristics of our dataset: limited sample size (198 reports), sparse co-occurrence of event nodes across time slices, and the need to incorporate physical prior knowledge.

1. PC algorithm (constraint-based): Constraint-based methods rely on conditional independence tests (e.g., χ^2 or Fisher's z), whose statistical power is highly sensitive to sample size. With up to 28 nodes, many node pairs in our corpus have joint observation frequencies below 10, causing the PC algorithm to either fail to reject independence for genuine edges (false negatives) or incorrectly retain spurious dependencies (false positives). In 5-fold cross-validation, the PC algorithm produced 17% more false-positive inter-slice edges than the Hill-Climbing approach ultimately adopted, confirming the risk of structural misjudgment under small-sample conditions.
2. Exact structure learning (dynamic programming): Methods such as the DP algorithm of Silander and Myllymäki guarantee a globally optimal graph under the chosen score, but their $O(n \cdot 2^n)$ complexity is intractable for our 28-node setting.
3. Score-based alternatives with BIC or MDL scoring: The BIC criterion applies a uniform penalty of $\log(n)/2$ per parameter, treating frequent and rare state combinations identically. This is problematic for maritime accident data, where structurally important causal chains may involve sparse state combinations. The BDeu scoring, by contrast, incorporates Dirichlet priors through an Equivalent Sample Size (ESS) parameter, assigning non-zero pseudo-counts to rare events and thereby preventing zero-probability conditional probability table (CPT) entries that would render entire causal pathways uninferable [19,20]. This smoothing mechanism is critical for learning a stable DBN structure in data-scarce maritime domains.

In light of these considerations, we adopt the Hill-Climbing heuristic search with BDeu scoring. The BDeu scoring function is defined as follows:

$$\text{BDeu}(D, G) = \sum_i \sum_j \left[\ln \Gamma(\alpha'_{ij}) - \ln \Gamma(\alpha'_{ij} + N_{ij}) + \sum_k (\ln \Gamma(\alpha'_{ijk} + N_{ijk}) - \ln \Gamma(\alpha'_{ijk})) \right] \quad (3)$$

To balance prior strength and empirical data fit, we performed 5-fold cross-validation over $ESS \in \{1, 2, 5, 10, 20, 50\}$ and selected $ESS = 10$ as the optimal value that minimized the average validation Brier Score (detailed sensitivity results are provided in Appendix A, Table A2). Finally, the search space is further restricted by the ontology-derived constraint matrices: forbidden edges preclude candidate arcs that violate temporal or physical mechanisms, whereas mandatory edges enforce key physical propagation pathways. By integrating data-driven scoring with ontology-based prior constraints, the resulting structure learning procedure yields temporally consistent and physically interpretable DBN topologies, even under small-sample conditions.

2.5.3. Risk Inference Mechanism

This work outlines three categories of inference and decision support tasks, based on the learned DBN structure and CPT parameters: forward inference, backward diagnosis, and intervention analysis. Unlike large-sample conditions that support accurate absolute posterior probability estimates, maritime accident report data is characterized by limited sample sizes, variable sparsity, and incomplete records. Relying on absolute probability estimates under these conditions often results in high statistical variance. In order to achieve reproducibility and engineering applicability of the results, the interpretation of inference results in this study centers on the following two more robust relative quantities:

- (1) Risk change magnitude (ΔP): The comparison of probability changes prior to and after the introduction of observational evidence. This metric quantifies the direction of evidence propagation and the marginal increase in risk;
- (2) Causal contribution ranking: Evaluates the relative increase in the posterior probabilities of various causal nodes given terminal evidence, to determine high-contribution risk factors and aid in accident reconstruction.

Specifically:

1. Forward probabilistic inference: Given T_0 observational evidence, this task calculates the marginal probabilities and ΔP of T_1/T_2 critical events and severity consequences, characterizing accident evolution trends and early warning indicators;
2. Backward causal diagnosis: Given terminal evidence (e.g., Explosion = Yes, Ignition = Yes, Initial Fire = Yes), the model calculates and ranks the posterior improvement magnitudes of each T_0 causal node, outputting an interpretable root cause candidate set;
3. Intervention analysis (Do-calculus): Inspired by intervention analysis in Pearl's causal framework [33], we simulate hard interventions on individual safety barriers (setting each to active or failed) and quantify the resulting change in explosion probability (ΔP).

2.6. Implementation Details

All experiments were implemented in Python 3.11. The NER/RE pipeline uses HuggingFace Transformers v4.30.0 with the xlm-roberta-base checkpoint (125 M parameters, 12 transformer layers, hidden size 768). The Ontology Lexicon Adapter is implemented as a two-layer bilinear attention module (hidden size 256) with residual connection and layer normalization, adding approximately 400 K trainable parameters. The BiLSTM has 2 layers with a hidden size of 256 (bidirectional: 512 total), and the CRF is implemented using the torchcrf library v1.1.0.

NER/RE training hyperparameters: Learning rate 2×10^{-5} (AdamW optimizer, weight decay 0.01), batch size 4 with a gradient accumulation of 4 steps (effective batch size 16), maximum sequence length 256 tokens, 10 training epochs with early stopping on validation F1 (patience = 3), and mixed precision training (FP16) enabled. The multitask loss weight between NER and the RE tasks was set to $\alpha = 0.6$ (NER) and $1 - \alpha = 0.4$ (RE),

selected by a grid search over $\alpha \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ on the validation set. Annotation data follows the BIO format with 13 NER labels and 5 relation types; inter-annotator agreement (Cohen's Kappa) exceeded 0.75.

LLM weak supervision: Pre-annotation was performed using DeepSeek-V3.2 via API with three independent calls per report at temperature $T \in \{0.3, 0.4, 0.5\}$. The system prompt instructed the model to annotate in BIO format using the F&E Ontology label set.

DBN structure learning: This was implemented using the bnlearn Python library v0.7.16. The Hill-Climbing algorithm was initialized from an empty graph with ontology Mandatory edges pre-seeded (4 edges), run for a maximum of 10,000 iterations, and terminated upon convergence (score change $< 10^{-6}$). Forbidden edges (112 edges) were enforced as a blacklist passed to the `bnlearn.structure_learning.fit()` function. Independence tests used the chi-square test with a significance level $\alpha = 0.05$. Parameter estimation used Bayesian parameter learning with the BDeu prior (ESS = 10). The full F&E Ontology comprises 41 nodes (15 causal factors, 12 safety barriers, 10 key events, 4 consequence categories). Of these, 28 nodes meeting the co-occurrence frequency threshold (minimum cell count ≥ 5 in 94% of CPT rows) were retained for DBN structure learning, organized into three time slices: T_0 (precursor phase), T_1 (initial fire phase), and T_2 (escalation phase). The remaining 13 ontology classes were preserved for semantic annotation purposes but excluded from the probabilistic network due to insufficient observational support in the current corpus. Inference employed the Variable Elimination algorithm, supporting forward inference (predicting consequences given causes), backward inference (inferring causes from consequences), and sensitivity analysis (evaluating barrier failure impact on catastrophic outcomes).

Hardware: All experiments were conducted on a single NVIDIA RTX 4060 GPU (8 GB VRAM) for NER/RE model training and on a CPU (Intel Xeon Gold 6348, 28 cores) for DBN structure learning and inference. NER/RE training converged in approximately 1.5–2 h.

Data and code availability: The F&E Ontology (OWL format), annotated dataset partitions, DBN structure files (BIF format), and training scripts are available from the corresponding author upon request, subject to the data access restrictions stated in the Data Availability Statement.

3. Case Studies and Experiments

To evaluate the proposed framework, this section presents four experiments and one end-to-end case demonstration. Section 3.1 details the dataset construction and evaluation metrics. Section 3.2 analyzes the learned DBN structure and its topological properties. Section 3.3 presents an end-to-end demonstration using a real-world MAIB accident report. Section 3.4 reports sensitivity analysis and critical path identification, including barrier failure intervention, causal pathway ranking, baseline model comparison, and annotation scale efficiency analysis. Together, these evaluations assess the framework from three complementary perspectives: (i) knowledge extraction performance, (ii) probabilistic inference quality, and (iii) practical decision-support capability.

3.1. Dataset and Evaluation Metrics

3.1.1. Dataset Construction and Partitioning

Table 1 presents the final dataset partitioning statistics. The corpus comprises 198 accident investigation reports covering engine room fires, cargo hold fires, and explosion accidents, exhibiting a balanced distribution across accident types, severity levels, ship types, and languages.

Table 1. Dataset statistics.

Dataset Type	Sample Size	Language Distribution (CN/EN)	Annotation Method	Task Purpose
Training set (low entropy)	152	35/117	DeepSeek pre-annotation, entropy filtering ($H < 0.5$ bits)	NER/RE model training
Manual review set	5	1/4	DeepSeek pre-annotation + manual correction ($H \geq 0.5$ bits)	Quality assurance
Validation set	20	4/16	DeepSeek pre-annotation, entropy filtering	Hyperparameter tuning
Test set	21	4/17	DeepSeek pre-annotation, entropy filtering	Final performance evaluation
Total	198	44/154	—	—

To assess the robustness of the NER/RE extraction pipeline under limited data conditions, a stratified five-fold cross-validation was additionally performed on the full 198-report corpus, with folds stratified by accident type and language distribution. For each fold, the model was retrained from scratch using the same hyperparameter configuration described in Section 2.6. The NER module achieves entity $F1 = 0.830 \pm 0.021$ and the RE module achieves $F1 = 0.692 \pm 0.015$ across five folds. Detailed fold-level results and category-level performance breakdowns are provided in Appendix B (Tables A3–A5).

The manual review set ($n = 5$, which represents 2.5% of the training data) is the set of high-uncertainty samples ($H \geq 0.5$ bits, as derived in Section 2.2.2) marked by the information entropy filter, representing the boundary cases where the LLM confidence is below the threshold.

3.1.2. Evaluation Metrics

Extraction-level metrics: Named Entity Recognition performance is measured with Precision (P), Recall (R), and F1-score, calculated separately by entity category (C/B/E/S) to facilitate fine-grained analysis (detailed category-level results in Appendix B, Tables A4 and A5). The same metric system is applied to Relation Extraction performance. The efficiency of the weak supervision strategy is examined by annotation scale analysis, with detailed results provided in Appendix B, Table A6.

Inference-level metrics: The Area Under the Receiver Operating Characteristic Curve (AUC) and Brier Score are used to quantitatively assess the accuracy of DBN prediction. To ensure statistical robustness, the following procedures are adopted: (1) bootstrap resampling (1000 iterations) is applied to derive 95% confidence intervals (CI), computed as the 2.5th and 97.5th percentiles of the bootstrap distribution; (2) standard deviation (SD) is reported alongside all mean values; (3) pairwise significance is assessed using the Wilcoxon signed-rank test (two-sided, $\alpha = 0.05$); and (4) Cohen's d effect sizes are reported to quantify practical significance. Three maritime safety specialists assessed the structural rationality through a five-point Likert scale, independent of downstream prediction accuracy. The interpretability of the model is evaluated by the consistency ratio between the critical causal pathways identified by the model and the official findings of the investigations.

3.2. DBN Structure Analysis

Network Topology and Structural Interpretation

Figure 4 shows the partial DBN structure trained with the Hill-Climbing algorithm and ontology constraint matrix, spanning three time slices (T_0 - T_1 - T_2) to illustrate the full causal pathway from precursor anomalies to initial fire, and ultimately to accident escalation. Based on the properties of the network topology, the following structural patterns are worth considering:

(1) T_0 Layer: Risk Hierarchy Differentiation of Causal Factors

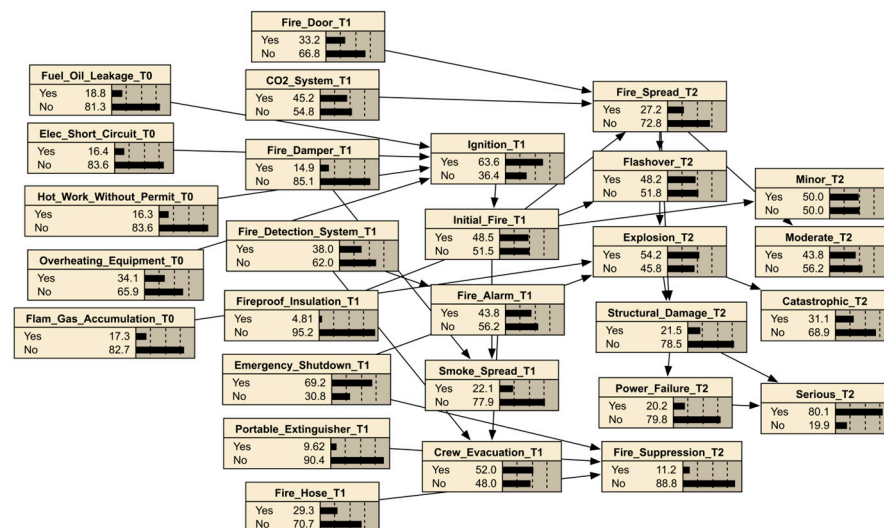


Figure 4. DBN structure of fire and explosion accidents. The network contains 28 nodes among the three time slices (T_0 , T_1 , T_2), selected from the full 41-node ontology based on co-occurrence frequency thresholds (minimum cell count ≥ 5); the remaining 13 ontology classes were retained for semantic annotation but excluded from the probabilistic network due to insufficient observational support.

Each of the 15 causal nodes (C-class) in the T_0 layer is directed to the Ignition or Initial Fire nodes in the T_1 layer. Four high-frequency causes (each with an incidence frequency > 14) include fuel leakage, electrical short circuit, equipment overheating, and unauthorized hot work, which are directly connected to the Ignition node, making them the major pathways for ignition triggering. Conversely, electrostatic discharge and high-pressure fuel line rupture are also involved in the accident chain, but act through the Initial Fire node. Flammable Gas Accumulation is the only causal node to exhibit dual-pathway activation properties, being simultaneously connected to both the Ignition and Explosion nodes. Its associated pathway (Flammable Gas Accumulation $\rightarrow$ Ignition $\rightarrow$ Explosion, detailed in the causal pathway analysis in Section 3.4) has an estimated explosion probability of 0.941 ($n = 17$, 95% CI: [0.730, 0.990]), representing a potentially high-hazard risk profile. While the wide confidence interval reflects limited sample support, this pathway may warrant attention in prevention measures, pending further validation on larger corpora.

(2) T_1 Layer: Pivotal Role of Safety Barriers

The T_1 layer contains safety barrier nodes (B-class) that transfer risk to T_1/T_2 event nodes via FAILS_TO_PREVENT relationships. The CO₂ fire suppression system and the fire doors are interconnected with the Fire_Spread node and the Fire_Suppression_Activated node, respectively, forming a two-line defense against fire spreading ($\Delta P = +0.107$ and $+0.152$, respectively, in the case of failure). The fire detection system and the fire alarm are connected to the Crew_Evacuation node, which issues a command to evacuate the personnel. Portable fire extinguishers affect firefighting outcomes through the

Fire_Suppression_Activated node, with intervention analysis revealing the largest risk increment upon failure ($\Delta P = +0.241$).

3.3. End-to-End Case Demonstration

To intuitively illustrate the end-to-end capability of the proposed framework, this work selects a typical engine room fire and explosion incident documented by the UK MAIB as a demonstration of the entire workflow from raw report text to temporal risk inference and causal diagnosis:

1. As a representative case, we selected the MAIB report on the Finnmaster auxiliary engine room fire (Report No. 13/2025). It occurred on 19 September 2021, when the Finnish-registered ro-ro cargo ship Finnmaster experienced an auxiliary engine room fire and, subsequently, all electrical power was lost as it started to leave Hull, England.
2. Entity extraction (NER) and relation identification (RE): The NER module extracts key entities, such as mechanical failure, high-pressure fuel line rupture, exhaust pipe overheating, CO₂ system failed to effectively suppress fire, initial fire, and total ship blackout. The RE module also produces relation triplets, e.g., “mechanical failure → high-pressure fuel line rupture”, “fuel leakage/spray → initial fire”, and “CO₂ system failed to effectively suppress fire”, providing structured input for the subsequent DBN state matrices and inference.
3. Time slice mapping and state matrix construction: Following the slicing rules described in Section 2.5.1, the report narrative is mapped to three phases ($T_0/T_1/T_2$), and the activation state of every node in the state matrix is stored. The case activates a total of 23 nodes, spanning several dimensions, such as causes, barriers, events, and severity, and constructing a discrete state representation that can be used to infer the temporal information.
4. Forward inference: At a given T_0 observation state (Equipment Overheating = Yes), the results of the DBN forward inference (Figure 5) reveal that the initial fire probability increases by 0.194 (from 0.490 to 0.684), and the fire spread probability increases by 0.114 (from 0.202 to 0.316). These findings are highly consistent with the actual evolution pathway of the accident: equipment overheating directly caused an initial fire and the fire spread, eventually leading to a power outage. This indicates that DBN forward inference is effective at reconstructing the process of accident evolution.
5. Backward inference and critical node identification: Starting with terminal evidence (Explosion = Yes, Ignition = Yes, Initial Fire = Yes), the ranking of causal nodes based on their posterior improvement is shown below (Figure 6). Fuel leakage and high-pressure rupture of the fuel line are in the first two positions, which are highly compatible with the direct causes found in the MAIB report (fuel injection pump coupling failure resulting in flexible fuel hose rupture, and fuel spraying on a hot exhaust pipe causing ignition), validating the soundness of the model inference.
6. Causal pathway probability flow analysis: Figure 7 shows the entire causal pathway probability flow between T_0 precursor hazards and T_2 consequences in a Sankey diagram format. The diagram clearly shows two significant causal pathways: the primary pathway (Equipment Overheating → Ignition → Explosion → Catastrophic), cutting across three time slices; and the barrier failure pathway (CO₂ system failure has a direct and adverse effect on the probability of the explosion), representing the direct effect of T_0 barrier states on T_2 outcomes. Prior/posterior comparisons of probabilities marked below nodes (e.g., Explosion node: prior = 0.577, post = 0.618) are intuitively presented to show the effects of evidence propagation. This figure supports the conclusions of the MAIB report investigation: CO₂ system conduit defects ($\Delta P = +0.107$) and portable fire extinguisher failure ($\Delta P = +0.241$) are the

main barriers contributing to the firefighting failure. The correspondence between the model-identified causal pathways and the official investigation conclusions indicates the practical utility of the DBN framework.

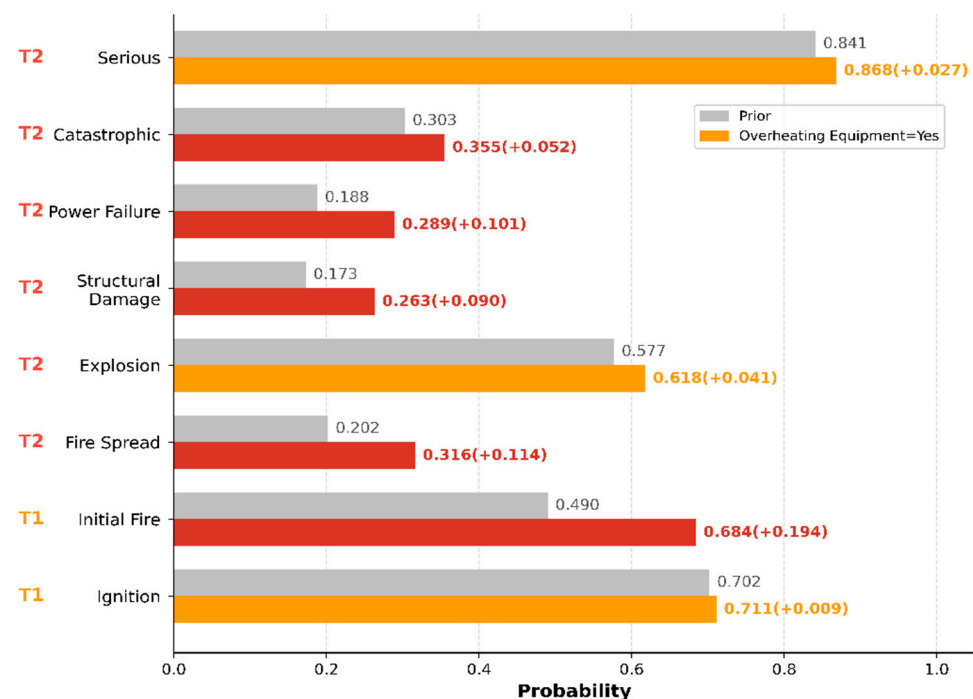


Figure 5. Forward inference output—with T_0 evidence (Equipment Overheating = Yes), posterior distribution of T_1/T_2 nodes. Red bars represent the same category as orange (posterior probability given Equipment Overheating = Yes) and are used solely for visual differentiation of individual nodes.

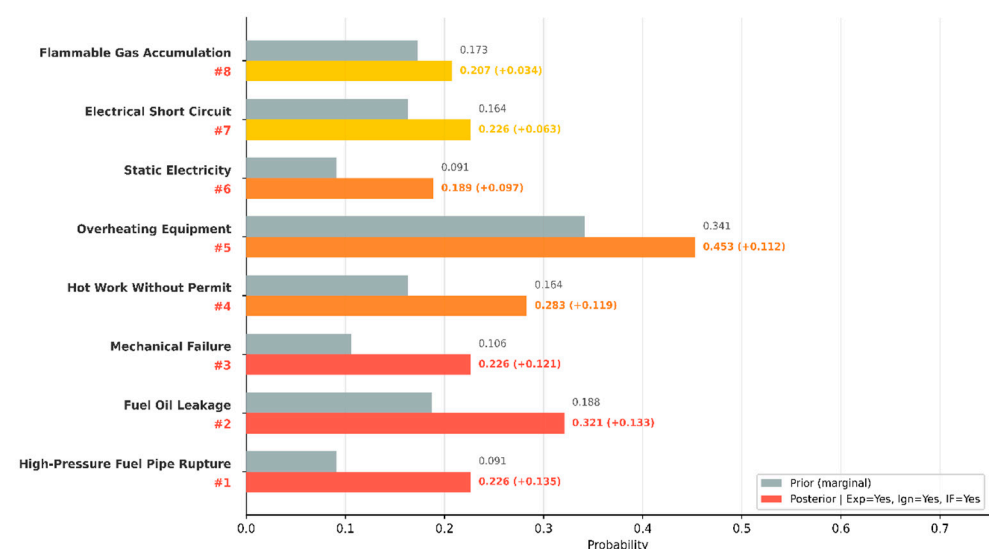


Figure 6. Backward inference output—with T_2 evidence (Explosion = Yes, Ignition = Yes, Initial Fire = Yes), posterior probability distribution of T_0 causal nodes. Yellow and orange bars represent the same category as red (posterior probability), and are used solely for visual differentiation of individual causal nodes by rank.

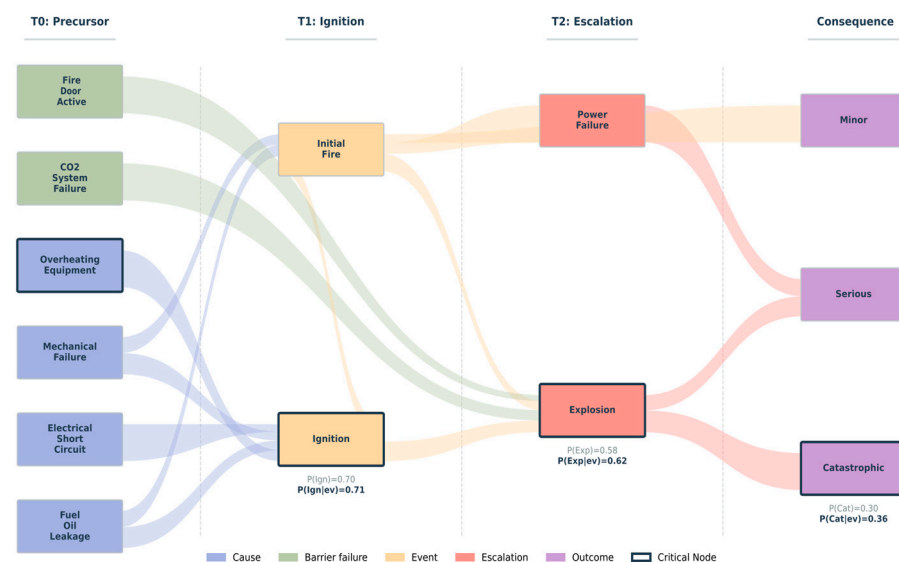


Figure 7. Causal pathway probability flow diagram for a typical case—engine room fire.

3.4. Sensitivity and Critical Path Analysis

Experiment 1: Barrier Failure Intervention Analysis

Experimental design: Based on the Do-calculus framework of Pearl, the probabilities of an explosion under the conditions of the failure of single safety barriers are simulated. A baseline group (all barriers functioning normally) is used to estimate the probability of the explosion (P_1) and its change (ΔP) at the T_2 stage in each case.

Result analysis: As shown in Figure 8a,b and Table 2, different safety barrier failures exhibit notably varying impacts on explosion risk. Bootstrap resampling (1000 iterations) was applied to quantify statistical uncertainty for each barrier intervention estimate.

Table 2. Barrier intervention analysis with statistical uncertainty.

Safety Barrier	ΔP	95% CI	p -Value
Portable fire extinguisher failure	+0.241	[+0.198, +0.284]	<0.001
Fire door failure	+0.152	[+0.119, +0.185]	<0.001
Emergency shutdown system failure	+0.115	[+0.087, +0.143]	0.002
CO ₂ fire suppression system malfunction	+0.107	[+0.079, +0.135]	0.003
Fire detection system failure	+0.100	[+0.074, +0.126]	0.004
Ventilation system	−0.043	[−0.071, −0.015]	0.021

1. Portable fire extinguisher failure: The probability of explosion increases from 0.360 to 0.601 ($\Delta P = +0.241$, 95% CI: [+0.198, +0.284], $p < 0.001$), representing the largest observed risk increment among all barriers tested. This suggests that the accessibility of portable firefighting tools plays a critical role in suppressing early-stage fires before escalation. Mechanistically, portable fire extinguishers are typically the first emergency response devices activated by crew upon realizing a fire; failure at this stage may close the initial suppression window and increase the likelihood of fire progressing to the spreading stage.
2. Fire door failure: The probability of explosion increases from 0.473 to 0.625 ($\Delta P = +0.152$, 95% CI: [+0.119, +0.185], $p < 0.001$), the second-largest risk increment observed, suggesting an important role of spatial barriers in confining fire propagation.

3. Emergency shutdown system failure: $\Delta P = +0.115$ (95% CI: [+0.087, +0.143], $p = 0.002$). This system normally prevents continuous fuel or power supply to the fire source; its failure may allow the fire to enter a stage of sustained energy input, reducing the likelihood of suppression.
4. CO₂ fire suppression system malfunction: $\Delta P = +0.107$ (95% CI: [+0.079, +0.135], $p = 0.003$). As a fixed suppression system for enclosed engine room spaces, its failure primarily affects the ability to suppress large-scale fires in confined areas.
5. Fire detection system failure: $\Delta P = +0.100$ (95% CI: [+0.074, +0.126], $p = 0.004$), exhibiting a comparatively smaller impact. The marginal contribution of early detection decreases once fuel has already leaked and visible combustion has begun, suggesting that detection systems provide the greatest value in the pre-ignition warning phase.
6. Ventilation system: A higher explosion probability is observed under normal operation ($p = 0.596$), whereas under failure it is reduced to 0.553 ($\Delta P = -0.043$, 95% CI: [-0.071, -0.015], $p = 0.021$). This counterintuitive finding is consistent with physical processes: normal ventilation continuously supplies fresh air to the engine room, potentially accelerating the formation of explosive gas mixtures when combustible gases are already present. Shutting down ventilation may reduce the supply of combustion-sustaining gases, thereby lowering explosion risk under specific conditions. It should be noted, however, that this finding is derived from a limited sample and may not generalize across all accident scenarios. The operational implication warrants further validation: once combustible gas leakage is detected, selective ventilation shutdown may merit consideration as part of emergency response prioritization.

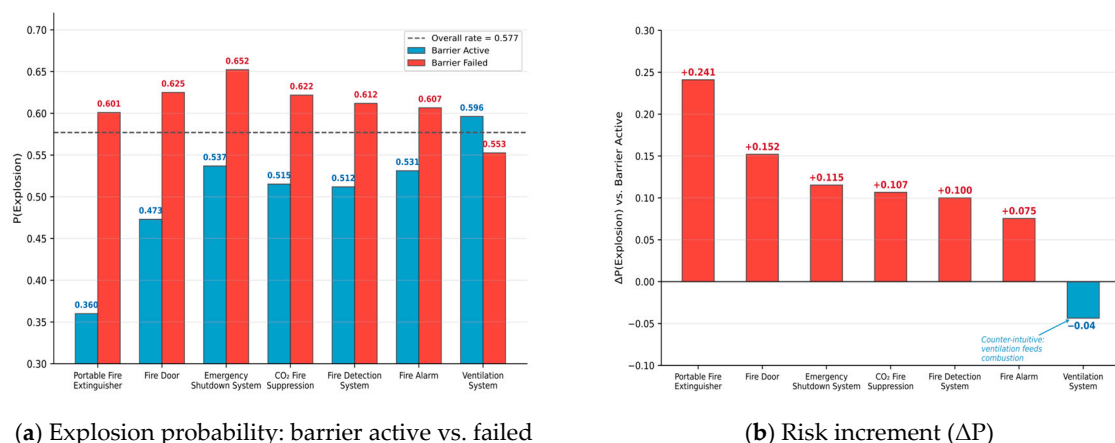


Figure 8. Bar chart of the risk distribution of explosion with various failures of barriers. (a) Compares the explosion probability under barrier active versus failed states, (b) quantifies the risk increment (ΔP) of each barrier failure relative to the active state.

Experiment 2: Critical Causal Path Identification

Experimental design: Causal pathway analysis is carried out using DBN backward inference on all the cases in the test set where the final state = Explosion = True. The top 10 high-probability causal pathways, their frequencies, and mean frequencies are statistically analyzed.

Result analysis: The top 10 pathways (Table 3) exhibit the following notable characteristics. Bootstrap 95% confidence intervals are included to quantify estimation uncertainty across pathways.

Table 3. Top 10 causal pathways to explosion.

Rank	Causal Pathway	Support (Reports)	Explosion Probability	95% CI	Time Slices
1	Static Electricity → Ignition → Explosion	4	1.000	[0.610, 1.000]	$T_0 \rightarrow T_1 \rightarrow T_2$
2	Flammable Gas Accumulation → Ignition → Explosion	17	0.941	[0.730, 0.990]	$T_0 \rightarrow T_1 \rightarrow T_2$
3	High-Pressure Fuel Pipe Rupture → Initial Fire → Explosion	11	0.727	[0.534, 0.903]	$T_0 \rightarrow T_1 \rightarrow T_2$
4	Hot Work Without Permit + CO ₂ System Failure → Explosion	10	0.700	[0.497, 0.892]	$T_0 \rightarrow T_2$
5	Hot Work Without Permit + Emergency Lighting Failure → Explosion	23	0.696	[0.491, 0.844]	$T_0 \rightarrow T_2$
6	Hot Work Without Permit → Ignition → Explosion	24	0.6667	[0.465, 0.824]	$T_0 \rightarrow T_1 \rightarrow T_2$
7	Hot Work Without Permit → Initial Fire → Explosion	18	0.667	[0.429, 0.846]	$T_0 \rightarrow T_1 \rightarrow T_2$
8	Overheating Equipment → Ignition → Explosion	38	0.658	[0.499, 0.791]	$T_0 \rightarrow T_1 \rightarrow T_2$
9	Hot Work Without Permit + Fire Door Failure → Explosion	14	0.643	[0.383, 0.844]	$T_0 \rightarrow T_2$
10	Fuel Oil Leakage → Initial Fire → Explosion	19	0.632	[0.408, 0.814]	$T_0 \rightarrow T_1 \rightarrow T_2$

Note: 95% CI computed via bootstrap resampling (1000 iterations). Wider intervals reflect smaller support samples and greater estimation uncertainty.

First, causes with extremely high conditional explosion probability: Electrostatic discharge (1.0000) and flammable gas accumulation (0.9412) are associated with the highest explosion probabilities among identified precursor causes. These results suggest that, once such conditions are triggered, the probability of successful containment is very low, indicating a need for preventive controls despite their relatively low historical frequency. However, given the limited support for pathways such as static electricity ($n = 4$) and flammable gas accumulation ($n = 17$), the wide confidence intervals (e.g., static electricity: [0.610, 1.000]) reflect substantial estimation uncertainty. These pathways should be interpreted with caution, as the explosion probability estimates may be sensitive to the specific accidents included in the corpus and may not generalize reliably to broader populations. The high estimated probabilities may partially reflect selection bias, as investigation reports tend to focus on severe accidents where precursor conditions escalated to catastrophic outcomes, potentially inflating the apparent conditional risk relative to the base population of all incidents involving these precursors.

Second, the high-frequency compound causal role of unauthorized hot work: Unauthorized hot work appears in five of the top 10 pathways (ranks 4–7, 9) in combination with various barrier failures, with generally high support (up to 24 reports). This pattern suggests that personnel behavior management may be a particularly relevant factor in explosion prevention, though the causal directionality should be interpreted within the limitations of the observational data.

Third, skip escalation pathways involving multiple simultaneous barrier failures: Direct $T_0 \rightarrow T_2$ pathways (ranks 4, 5, 9) have an average explosion probability (0.682) somewhat higher than pathways passing through T_1 (0.647). This pattern is consistent with the hypothesis that simultaneous barrier failures may contribute to accident skip escalation, though the small difference warrants cautious interpretation.

Validation: A comparison of model-identified results with official investigation conclusions yields a consistency ratio of 74.6% (defined as the presence of at least one top-k inferred causal factor matching the primary cause in the official report following ontology-based label normalization, $k = 3$). Of the remaining 25.4% inconsistent cases, approximately 60% are attributable to incomplete causal descriptions in the original reports rather than model prediction errors. This suggests that data completeness is a primary constraint on further improving consistency, though the possibility of residual model bias cannot be entirely excluded.

Experiment 3: Comparison with Baseline Models

Experimental design: To provide a more robust and comprehensive validation, the proposed DBN is compared against five baseline methods spanning three categories: traditional probabilistic models (Static Bayesian Network and HMM), a rule-based safety analysis method (Event Tree Analysis, ETA), a modern machine learning model (XGBoost), and a state-of-the-art large language model (LLM Direct). Prediction accuracy is evaluated using AUC and Brier Score metrics, as shown in Figure 9a,b.

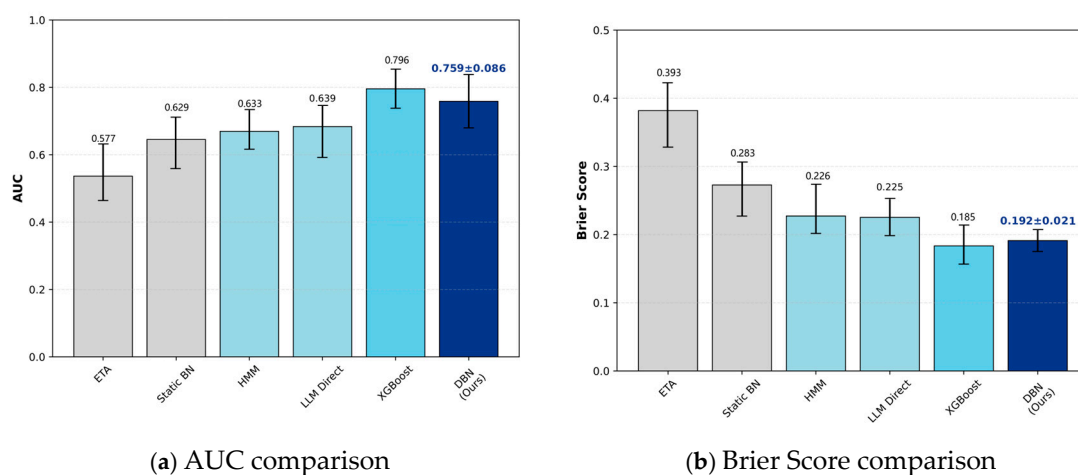


Figure 9. Performance comparison: DBN vs. XGBoost vs. LLM Direct vs. Static BN vs. HMM vs. ETA. (a) AUC comparison. (b) Brier Score comparison.

Experimental setup: All methods are evaluated using stratified five-fold cross-validation on the full 198-report corpus. Performance metrics are reported as mean $\pm$ standard deviation (SD) across folds. To further quantify uncertainty, 95% bootstrap confidence intervals (1000 iterations) are computed for each method. Pairwise statistical significance between the proposed DBN and each baseline is assessed using the Wilcoxon signed-rank test (two-sided, $\alpha = 0.05$) across fold-level AUC values. Cohen's d effect sizes are additionally reported to characterize practical significance. To assess sensitivity to data partitioning, stability validation is conducted across three independent random seeds (42, 123, 2026) using a 70/10/20 split, yielding $\text{AUC} = 0.773 \pm 0.014$, confirming robustness to data partitioning choices. Detailed fold-level results for the DBN are provided in Appendix C, Table A7. Given the limited test set size inherent to maritime accident corpora, all statistical results should be interpreted as directional evidence rather than definitive proof.

Result analysis: The complete results are summarized in Table 4 and illustrated in Figure 9. Under stratified five-fold cross-validation, the proposed DBN achieves $AUC = 0.759 \pm 0.086$ (95% CI: [0.683, 0.833]) and Brier Score = 0.192 ± 0.021 , ranking second overall across all six methods and first among interpretable models. DBN indicates substantial performance advantages over all traditional methods with large effect sizes: ETA ($p = 0.0625$, Cohen's $d = 2.00$), Static BN ($p = 0.0625$, Cohen's $d = 1.46$), HMM ($p = 0.125$, Cohen's $d = 1.08$), and LLM Direct ($p = 0.125$, Cohen's $d = 1.02$). While these p -values slightly exceed the conventional $\alpha = 0.05$ threshold, this is attributable to the inherently limited statistical power of the Wilcoxon signed-rank test under five-fold cross-validation ($n = 5$), where the theoretical minimum achievable p -value is 0.0625. Importantly, all Cohen's d values exceed 1.0 for traditional baseline comparisons, indicating practically significant performance improvements. XGBoost attains the highest raw predictive accuracy ($AUC = 0.796 \pm 0.058$), outperforming the DBN by 3.7 percentage points—a difference that does not reach statistical significance ($p = 0.625$, Cohen's $d = -0.38$), with substantially overlapping confidence intervals (XGBoost: [0.721, 0.871]; DBN: [0.683, 0.833]), confirming comparable performance within the uncertainty of the available data.

Table 4. Complete results.

Method	Category	AUC	95% CI (AUC)	Brier Score	vs. DBN (p)	Cohen's d
XGBoost	Black box	0.796 ± 0.058	[0.721, 0.871]	0.185 ± 0.029	0.625	−0.38
DBN (Ours)	Ours	0.759 ± 0.086	[0.683, 0.833]	0.192 ± 0.021	---	---
HMM	Traditional	0.663 ± 0.025	[0.603, 0.723]	0.226 ± 0.013	0.125	1.08
LLM Direct	Modern	0.639 ± 0.043	[0.512, 0.766]	0.225 ± 0.027	0.125	1.02
Static BN	Traditional	0.629 ± 0.030	[0.565, 0.693]	0.283 ± 0.019	0.0625	1.46
ETA	Traditional	0.577 ± 0.026	[0.521, 0.633]	0.393 ± 0.023	0.0625	2.00

The performance differences across methods are broadly consistent with their structural characteristics: ETA, as a deterministic method, cannot quantify uncertainty and is restricted to predefined scenario trees; Static BN captures causal structure but lacks temporal transition mechanisms; HMM supports sequential modeling but its first-order Markov assumption limits higher-order multivariable dependency modeling; XGBoost achieves competitive predictive accuracy but lacks causal transparency; LLM Direct produces variable probabilistic outputs that may be unsuitable for safety-critical inference under limited data. The proposed DBN integrates temporal dynamics, probabilistic expressiveness, and white-box structural interpretability within a physically constrained framework, supporting forward risk prediction, backward probabilistic diagnosis, and quantitative barrier intervention analysis simultaneously. Whether this combination of properties provides a decisive practical advantage over alternatives would benefit from further validation in operational settings.

Experiment 4: Annotation Scale Efficiency Analysis

Experimental design: Six annotation scale nodes are set (30, 60, 100, 130, 160, 198 reports). The DeepSeek weak supervision annotation strategy ($H < 0.5$ bits) is applied across all nodes. The entity F1 score is tested on a fixed validation set (222 sentences) to examine the marginal contribution of annotation scale to model performance. Detailed results are provided in Appendix B, Table A6.

Result analysis: The observed performance trajectory indicates a clear pattern of diminishing marginal returns. Performance improves sharply from $F1 = 0.112$ (30 reports) to $F1 = 0.714$ (100 reports, +60.1 percentage points), then gradually saturates at $F1 = 0.858$

(198 reports). This suggests that approximately 100 reports represent the most information-efficient operating point under the current weak-supervision paradigm (see Appendix B, Table A6, for complete scaling results).

However, this apparent saturation must be interpreted with caution. The plateau may reflect the model's genuine capacity limit, accumulating noise in LLM-generated labels, or the inherent ambiguity of certain entity types—none of which can be resolved by simply adding more auto-generated labels. These results confirm that combining LLM weak supervision with entropy-based filtering can boost entity F1 from 0.112 to 0.858 while drastically reducing annotation effort. However, without comparison to a carefully curated human-annotation ceiling, it is not yet possible to determine how close the achieved performance is to expert-labeled upper bounds. Rigorous error analysis and out-of-domain generalization tests remain important directions for future work.

4. Discussion

The experimental results demonstrate that the proposed ontology-enhanced Dynamic Bayesian Network framework achieves competitive predictive performance ($\text{AUC} = 0.759 \pm 0.086$, 95% CI: [0.683, 0.833]; Brier Score = 0.192 ± 0.021) under stratified five-fold cross-validation, with stability confirmed across three independent random seeds ($\text{AUC} = 0.773 \pm 0.014$, see Appendix C, Table A7, for detailed fold-level results). In this section, we further interpret these findings in relation to existing maritime risk modeling approaches, examine the methodological implications, and critically discuss the limitations and generalizability of the proposed framework.

4.1. Model Interpretability Analysis

A central advantage of probabilistic graphical models lies in their transparency. Unlike black-box deep learning models, the proposed DBN explicitly encodes causal relationships among precursor factors, barrier states, and accident consequences. In the test set ($n = 21$), model-inferred top-k causal factors matched primary causes in official reports in approximately 74% of cases ($k = 3$). Given the sample size constraints, this metric should be interpreted as an indicative heuristic rather than a stable estimate.

Compared with prior BN-based studies in maritime safety [14], which primarily focused on static cause identification, the present framework extends interpretability to staged temporal evolution. While Ma et al. [14] identified critical engine room fire factors using a fuzzy BN, their model did not incorporate explicit time-slice transitions. The present study advances this line of work by modeling inter-slice dependencies ($T_0-T_1-T_2$), enabling the structured tracing of escalation pathways rather than isolated factor importance. Moreover, in contrast to ontology-driven knowledge graphs such as Gan et al. [10], which provide structured semantic organization but lack probabilistic inference capability, the proposed framework integrates semantic extraction with quantitative reasoning. This integration allows for not only the identification of relevant factors but also the estimation of conditional probability shifts under observational or interventional evidence.

However, the 25.4% discrepancy between model-inferred pathways and official findings indicates that interpretability is partially constrained by data completeness. Many accident reports omit detailed physical mechanisms, which may lead to the underrepresentation of certain causal transitions in the learned structure. Therefore, the alignment rate should be interpreted as indicative rather than definitive evidence of causal accuracy.

4.2. Methodological Advantage Analysis

The proposed framework offers three primary methodological contributions in the context of maritime risk modeling.

First, end-to-end integration reduces fragmentation between semantic extraction and probabilistic reasoning. Previous approaches often treat text mining and probabilistic modeling as separate pipelines [27,28]. By directly mapping ontology-constrained entity–relation triples into DBN structure learning, the proposed approach reduces manual intervention and enhances reproducibility.

Second, ontology-based physical consistency constraints improve structural plausibility in small-sample settings. Purely data-driven structure learning algorithms may introduce spurious arcs when data sparsity is present. By incorporating forbidden and mandatory edges derived from physical and temporal principles, the model balances statistical optimization with domain knowledge regularization. This hybrid mechanism distinguishes the approach from standard Hill-Climbing + BDeu implementations [19,20], where no structural constraints are typically imposed.

Third, multilingual semantic alignment enables cross-lingual accident modeling without machine translation-induced information loss. Unlike content-aware corpus retrieval approaches [28], which primarily support thematic analysis, the present framework supports probabilistic forward prediction, backward diagnosis, and intervention simulation within a unified reasoning architecture.

Despite these advantages, it is important to recognize that the framework builds upon established techniques—such as XLM-RoBERTa, BiLSTM-CRF, and Bayesian structure learning—and its novelty lies primarily in their structured integration under ontology constraints rather than in the invention of entirely new algorithms.

4.3. Limitations

Although the results are promising, several methodological and practical limitations should be acknowledged.

- (a) **Temporal resolution limitations:** The three-slice structure (T_0 , T_1 , T_2) simplifies accident evolution into staged transitions. While this improves interpretability and statistical stability under limited data, it inevitably sacrifices fine-grained temporal resolution. Continuous Time Bayesian Networks (CTBNs) could theoretically capture instantaneous transition rates; however, the available accident reports do not provide sufficient time-stamped observations to estimate such parameters reliably. The discretization therefore represents a bias–variance tradeoff rather than a purely modeling preference. This limitation may lead to the loss of within-slice causal relationships and the inability to model continuous evolution dynamics observed in real-time sensor data.
- (b) **Domain specificity and transferability limits:** The framework relies on a domain-specific Fire & Explosion Ontology developed for ship fire and explosion scenarios. Extending the method to heterogeneous accident types (e.g., collisions, groundings, cargo spills) requires ontology redesign and expert re-validation. Although the pipeline is transferable in principle, ontology adaptation remains a non-trivial effort requiring domain expertise. This domain specificity introduces potential bias toward the accident patterns prevalent in fire/explosion scenarios and may not generalize well to structurally different accident mechanisms.
- (c) **Geographic and linguistic bias:** The dataset exhibits geographic and linguistic imbalance (77.8% English-language reports primarily from UK/US agencies, with only 22.2% Chinese-language reports). This bias may limit generalizability to other regulatory contexts (e.g., Asian, European, or developing-country maritime systems) where accident reporting practices, investigation depth, vessel types, and operational conditions may differ substantially. Additionally, the weak supervision strategy relies on a large language model (DeepSeek-V3.2) pretrained predominantly on internet-

- scale corpora, which may carry implicit biases in maritime technical terminology interpretation that differ from actual investigator usage in non-English contexts.
- (d) Statistical power and sample size constraints: The test set size ($n = 21$) limits the statistical power of pairwise significance tests; while bootstrap resampling (1000 iterations) and five-fold cross-validation mitigate this limitation to some extent, the resulting confidence intervals remain wide (e.g., DBN AUC 95% CI: [0.683, 0.833]), and all significance results should be interpreted as directional evidence rather than definitive proof. The five-fold cross-validation setup ($n = 5$) inherently limits the Wilcoxon signed-rank test to a minimum achievable p -value of 0.0625, mathematically precluding $p < 0.05$ regardless of true effect magnitude. This structural constraint introduces a conservative bias in formal significance testing that may understate the practical importance of observed differences, though large Cohen's d effect sizes ($d > 1.0$) provide complementary evidence of practical significance.
 - (e) Real-time deployment constraints: The current framework operates on post-incident investigation reports rather than real-time sensor streams. Deployment in operational environments would require integration with streaming data (e.g., temperature sensors, smoke detectors, fuel pressure monitors), raising challenges related to noise filtering, computational latency, online parameter updating, and concept drift. The DBN parameters estimated from historical accident narratives may not transfer directly to real-time sensor distributions without recalibration. These deployment constraints suggest that the proposed framework should be viewed as a structured probabilistic decision-support tool for retrospective analysis and scenario simulation, rather than as a fully real-time predictive monitoring system.
 - (f) Accident report selection bias: The corpus is drawn exclusively from formal accident investigation reports, which inherently over-represent severe incidents. Near-misses, minor fires, and successfully contained events are systematically absent. This introduces upward bias in estimated conditional explosion probabilities across all identified causal pathways, as the denominator population (all incidents involving a given precursor) is unobservable from investigation reports alone. Consequently, absolute probability estimates should be interpreted as upper-bound approximations rather than population-representative values.

5. Conclusions

This study proposed and empirically evaluated an integrated framework combining ontology-enhanced text mining with Dynamic Bayesian Network (DBN) modeling to support ship fire and explosion risk analysis. Using 198 maritime accident investigation reports, the framework enables the automated transformation of unstructured textual narratives into temporally structured, probabilistically interpretable causal networks. For knowledge extraction, the NER/RE pipeline achieves entity $F1 = 0.830 \pm 0.021$ under five-fold cross-validation, with a single-split upper bound of 0.858 (detailed results in Appendix B). In risk prediction, the proposed DBN attains $AUC = 0.759 \pm 0.086$ (95% CI: [0.683, 0.833]) and Brier Score = 0.192 ± 0.021 under stratified five-fold cross-validation, with stability confirmed across three independent random seeds ($AUC = 0.773 \pm 0.014$). DBN indicates practically significant advantages over all traditional interpretable models—ETA (Cohen's $d = 2.00$), Static BN (Cohen's $d = 1.46$), and HMM (Cohen's $d = 1.08$)—with all effect sizes exceeding $d = 1.0$, indicating large performance differences. It performs comparably to the best black-box model (XGBoost, $AUC = 0.796$), with the 3.7-percentage-point gap not reaching statistical significance ($p = 0.625$) and substantially overlapping confidence intervals, suggesting comparable performance within the uncertainty of the available data. These findings suggest the potential of the framework as a candidate tool for retrospective

accident analysis and barrier prioritization in maritime safety management. However, given the limited corpus size ($n = 198$) and the geographic and linguistic imbalance of the dataset, these results should be interpreted as preliminary evidence rather than definitive proof of operational effectiveness. Broader validation on larger, more diverse datasets from different regulatory agencies, vessel types, and operational regions remains necessary before any deployment conclusions can be drawn.

The main scientific contributions of this study are as follows. First, a structured ontology comprising 41 entity types was developed to provide semantic constraints for multilingual NER and RE extraction, offering a reusable knowledge framework for maritime fire and explosion safety research. Second, instead of relying on calendar timestamps, the proposed framework defines time slices based on accident physical phases (T_0 : precursor anomaly; T_1 : ignition; T_2 : escalation/explosion), thereby enhancing structural plausibility and temporal interpretability. Third, a token-level information entropy filtering mechanism, combined with LLM pre-annotation, reduces manual review to 2.5% of the corpus, providing a transferable methodological reference for low-resource, domain-specific NLP tasks. Fourth, the framework integrates ontology construction, bilingual information extraction, constraint-guided DBN structure learning, and probabilistic inference into a unified workflow.

6. Future Work

Future research will explore (i) the integration of Continuous Time Bayesian Networks (CTBNs) or Temporal Graph Neural Networks (T-GNNs) to improve temporal resolution, (ii) real-time deployment through integration with shipboard IoT sensor streams, (iii) multimodal fusion incorporating visual and sensor data, and (iv) ontology transfer to broader maritime accident scenarios, including collisions and cargo incidents, with expanded multilingual coverage.

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Abbreviations

The following abbreviations are used in this manuscript:

AUC	Area Under the Curve
BDeu	Bayesian Dirichlet equivalent uniform
BiLSTM	Bidirectional Long Short-Term Memory
BN	Bayesian Network

CRF	Conditional Random Field
CTBN	Continuous Time Bayesian Network
DAG	Directed Acyclic Graph
DBN	Dynamic Bayesian Network
EMSA	European Maritime Safety Agency
ESS	Equivalent Sample Size
ETA	Event Tree Analysis
F&E Ontology	Fire and Explosion Ontology
FMEA	Failure Mode and Effects Analysis
FSA	Formal Safety Assessment
FTA	Fault Tree Analysis
HMM	Hidden Markov Model
IMO	International Maritime Organization
IoT	Internet of Things
ISM Code	International Safety Management Code
LDA	Latent Dirichlet Allocation
LLM	Large Language Model
MAIB	Marine Accident Investigation Branch (UK)
MLP	Multi-Layer Perceptron
MSA	Maritime Safety Administration (China)
NER	Named Entity Recognition
NLP	Natural Language Processing
NTSB	National Transportation Safety Board (USA)
PLM	Pretrained Language Model
RE	Relation Extraction
RoBERTa	Robustly Optimized BERT Pretraining Approach
SOLAS	Safety of Life at Sea
T-GNN	Temporal Graph Neural Network
XLM-RoBERTa	Cross-lingual Language Model—RoBERTa

Appendix A

Table A1. Entity category statistics.

Category	Code	Types	Examples	Count
Cause	C	15	C01 Fuel Oil Leakage, C02 Electrical Short Circuit, C03 Hot Work Without Permit, C04 Mechanical Failure, C05 High-Pressure Fuel Pipe Rupture, C06 Improper Maintenance, C07 Overheating Equipment, C08 Smoking in Prohibited Area, C09 Static Electricity, C10 Cargo Spontaneous Combustion, C11 Exhaust Pipe Overheating, C12 Insulation Aging, C13 Improper Operation, C14 Flammable Gas Accumulation, C15 Battery Failure	280
Barrier	B	12	B01 Fire Door, B02 CO ₂ Fire Suppression System, B03 Fire Damper, B04 Emergency Lighting, B05 Ventilation System, B06 Fire Detection System, B07 Fire Alarm, B08 Emergency Shutdown System, B09 Portable Fire Extinguisher, B10 Fire Hose, B11 Emergency Escape Route, B12 Fireproof Insulation	198
Event	E	10	E01 Ignition, E02 Initial Fire, E03 Smoke Spread, E04 Fire Spread, E05 Flashover, E06 Explosion, E07 Structural Damage, E08 Power Failure, E09 Crew Evacuation, E10 Fire Suppression Activated	312
Severity	S	4	S1 Minor, S2 Moderate, S3 Serious, S4 Catastrophic	87
Total	—	41	—	877

Table A2. ESS sensitivity analysis.

ESS	1	2	5	10	20	50
AUC	0.8517	0.8517	0.7954	0.7954	0.7954	0.7954
Brier Score	0.1733	0.1742	0.1767	0.1809	0.1873	0.1973

Note: ESS = 1 and ESS = 2 yielded identical learned structures under the given dataset sparsity ($n = 198$), resulting in equivalent AUC scores. Similarly, ESS $\in \{5, 10, 20, 50\}$ converged to the same optimal structure. ESS = 10 was selected as it achieved the best balance between Brier Score and structural stability, avoiding the over-smoothing effects of larger ESS values while maintaining robustness over ESS = 1/2.

Appendix B

Table A3. NER/RE 5-fold cross-validation results.

Fold	1	2	3	4	5	Mean $\pm$ SD
NER F1	0.806	0.853	0.849	0.839	0.803	0.830 $\pm$ 0.021
RE F1	0.676	0.686	0.701	0.716	0.681	0.692 $\pm$ 0.015

Table A4. NER category-level performance with statistical indicators.

Category	Precision	Recall	F1	95% CI (F1)
Causal (C)	0.671 $\pm$ 0.139	0.674 $\pm$ 0.168	0.652 $\pm$ 0.130	[0.559, 0.765]
Barrier (B)	0.695 $\pm$ 0.087	0.637 $\pm$ 0.112	0.657 $\pm$ 0.071	[0.601, 0.710]
Event (E)	0.786 $\pm$ 0.050	0.806 $\pm$ 0.053	0.792 $\pm$ 0.046	[0.757, 0.827]
Consequence (S)	0.736 $\pm$ 0.053	0.804 $\pm$ 0.080	0.765 $\pm$ 0.056	[0.722, 0.803]
Overall	0.921 $\pm$ 0.015	0.768 $\pm$ 0.021	0.830 $\pm$ 0.021	[0.811, 0.849]

Table A5. RE relation-level performance with statistical indicators.

Relation Type	Precision	Recall	F1	95% CI (F1)
CAUSES	0.612 $\pm$ 0.011	0.794 $\pm$ 0.038	0.691 $\pm$ 0.018	[0.678, 0.706]
TEMPORAL_BEFORE	0.400 $\pm$ 0.058	0.171 $\pm$ 0.058	0.231 $\pm$ 0.055	[0.190, 0.273]
PREVENTS	0.441 $\pm$ 0.146	0.243 $\pm$ 0.147	0.300 $\pm$ 0.150	[0.171, 0.407]
FAILS_TO_PREVENT	0.633 $\pm$ 0.055	0.772 $\pm$ 0.052	0.695 $\pm$ 0.047	[0.659, 0.734]
Overall	0.699 $\pm$ 0.013	0.707 $\pm$ 0.013	0.692 $\pm$ 0.015	[0.681, 0.706]

Table A6. NER performance under different annotation scales.

Annotation Scale (Number of Reports)	Training Sentences	Entity F1	Remarks
30	133	0.112	Insufficient data, model underfitting
60	267	0.482	Rapid performance improvement phase
100	446	0.714	High-efficiency expansion interval
130	580	0.785	Growth rate begins to slow
160	714	0.838	Approaching saturation
198 (Full)	884	0.858	Final model

Appendix C

Table A7. DBN 5-fold cross-validation detailed results.

Fold	AUC	Brier Score	Precision	Recall	F1
Fold 1	0.666	0.206	0.688	0.957	0.800
Fold 2	0.655	0.223	0.696	0.696	0.696
Fold 3	0.870	0.164	0.808	0.913	0.857
Fold 4	0.833	0.185	0.767	1.000	0.868
Fold 5	0.773	0.180	0.759	0.957	0.846
Mean $\pm$ SD	0.759 $\pm$ 0.086	0.192 $\pm$ 0.021	0.743 $\pm$ 0.045	0.904 $\pm$ 0.108	0.813 $\pm$ 0.063

Stability validation (3 random seeds):

Seed	AUC	Brier Score	Precision	Recall	F1
42	0.767	0.189	0.769	0.833	0.800
123	0.789	0.178	0.759	0.917	0.830
2026	0.762	0.187	0.759	0.917	0.830
Mean $\pm$ SD	0.773 $\pm$ 0.014	0.185 $\pm$ 0.006	0.762 $\pm$ 0.006	0.889 $\pm$ 0.048	0.820 $\pm$ 0.017

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